

The Returns to Schooling and Experience

Introduction

Does additional education or work experience boost earnings? These relationships are of great interest to both researchers and policymakers. Education represents a significant investment by both students (in time and money) and society (through public funding of education), so understanding how education relates to earnings is a key public concern. Basic economic theory suggests that earnings should be positively affected by additional schooling. Ostensibly, more educated individuals accumulate greater human capital via increased skills, training, knowledge, etc. Consequently, in equilibrium where workers are paid their marginal productivity, higher educated workers get paid more. Similarly, work experience should also be positively related to earnings. In this paper, I use Census data to explore the validity of this theory, assessing visual and analytical evidence to determine whether education or work experience does indeed correspond to more earnings.

Data Exploration

The data used in this paper comes from the 2017 sample of the American Community Survey (ACS), accessed via IPUMS. This data is a 1-in-100 national random (weighted) sample of the population. The ACS collects Census data on housing and population at the individual-level including information on employment, income, education, and various demographic and household characteristics.

For my analysis, the key variables I need are labor income, years of schooling, and years of work experience. The ACS includes several income variables including wage and salary income, which I will use to measure earnings. This variable corresponds to total wage income in the previous year of the sample. Notably, ACS codes missing values for wage and salary income as 999999 and 999998, so I drop these values from my sample. Additionally, I drop any observations with non-positive earnings from my sample because negative wage income should not be possible as such values likely represent errors of some sort.

I define years of schooling using ACS's education variable, which bins years of schooling as nursery school to grade 4, grades 5 through 8, grades 9 - 16 binned separately, and then 5+ years of college. Notably, very few persons report education attainment less than a high school diploma. In order to avoid bias from these outlier values, I drop any observations with less than twelve years of schooling and define my schooling variable as years of schooling beyond a high school diploma (or 12 years of schooling). For example, someone with four years of college would have 4 years of schooling beyond high school and a high school graduate would have zero years of schooling beyond high school. Notably, I do not want to include persons still in school since their education is ongoing. Therefore, I limit my sample to respondents that have completed their schooling.

The ACS does not include a variable for years of work experience; however, I impute this by making two assumptions. First, I assume that everyone completes their schooling on a similar timeline such that high school graduates are 18, 4-year college graduates are 22, etc. Second, I assume that everyone enters the workforce immediately upon finishing their school and stays in the workforce continually thereafter. In other words, I effectively study the years of potential experience. Consequently, I define years of work experience as a person's age minus their years of schooling beyond high school plus eighteen. For a high school graduate, years of work is equal to their age minus eighteen. Notably, there are a few odd observations where these assumptions can be seen to not hold up since work experience is negative, so I drop these from my sample. There are some important limitations to defining work experience this way, which I'll discuss at greater length in the analysis section.

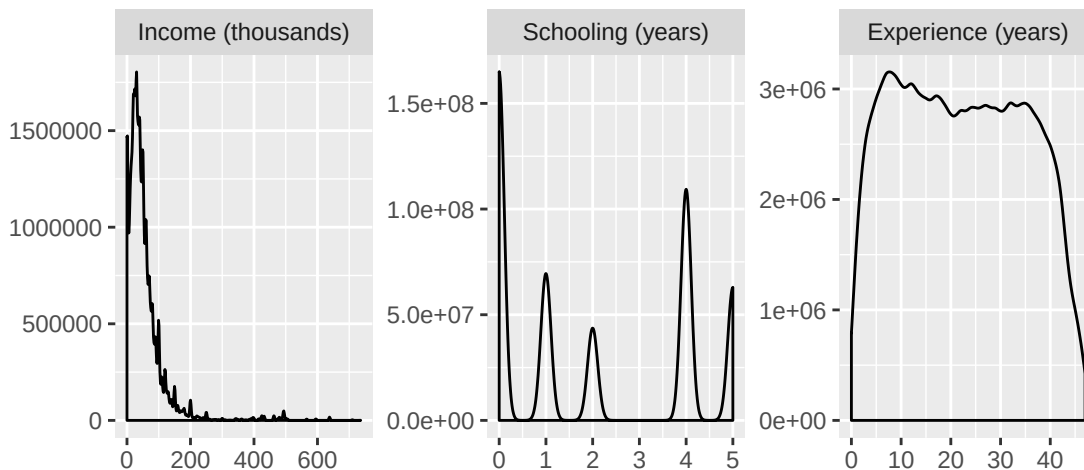
I limit my sample to 18-65 year old persons to focus on the prime age workforce and avoid biasing my results from extreme observations that are either very young or very old. Finally, I limit my sample to only persons who worked full-time (i.e., 50-52 weeks), which makes my previous assumption that potential years of work is equivalent to actual years of work more tenable. The various restrictions I place on the data reduce my

Table 1: Summary Statistics

	Mean	Median	SD	Min	Max	P25	P75	N
Income (thouands)	54.137	40	62.759	0	736	20	67	123068841
Schooling	2.019	1	1.962	0	5	0	4	123068841
Experience	22.131	22	12.517	0	47	11	33	123068841

sample size considerably, but these restrictions suggest the individuals left in my sample have strong labor force attachment, and are most relevant to my study.

Figure 1



The distributions of the three key variables (i.e., earnings, years of schooling, and work experience) are presented in Figure 1 and summary statistics are given in Table 1. The typical worker earns roughly forty thousand dollars. Notably, the mean is quite a bit higher than the median due to a large number of extreme outlier values in the right tail of the distribution. Most workers earn between twenty and seventy thousand dollars annually. Ostensibly, the two most common schooling outcomes are high school graduate and four years of college. The distribution of work experience is roughly uniform. All in all, these stylized facts seem sensible given the definitions and restrictions outlined above.

Data Analysis

Are education or experience positively associated with higher earnings? As a first pass at answering this question, I'll stratify the data by education and experience. Specifically, I'll split the data into High/Low Education or Experience based on whether an individual is above or below the median. Having done this, I can examine summary statistics of income conditional on these groupings. Table 2 shows summary statistics of income grouped by education and table 3 shows summary statistics of income grouped by experience.

Ostensibly, the typical worker with low schooling (i.e., high school graduate only) earns considerably less than the typical worker with high schooling. The difference is roughly six thousand dollars. Similarly, the interquartile range of incomes is higher for high education workers. This result is in-line with my expectations since education should increase a worker's productivity, and therefore their earnings. My findings are similar comparing income and work experience. The typical worker with low experience (i.e., less than about 20 years) earns about 10 thousand dollars less than the typical worker with high experience. In addition, the interquartile range is slightly broader and shifted upward for high experience workers. These patterns are moreover reflected in Figure 2, which plots histograms of earnings conditional on education or experience

Table 2: Summary Statistics of Income

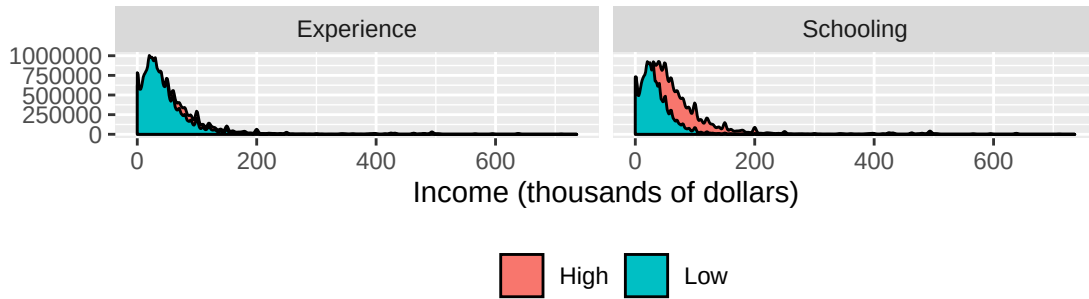
Schooling	Mean	Median	SD	Min	Max	P25	P75	N
Low	34.853	28.8	34.722	0	736	15	45	45041608
High	65.268	50.0	71.956	0	736	26	80	78027233

Table 3: Summary Statistics of Income

Experience	Mean	Median	SD	Min	Max	P25	P75	N
Low	47.263	35	52.64	0	736	20	60	60440275
High	60.770	45	70.54	0	736	24	75	62628566

groups. Overall, these findings indicate that both education and experience may be positively associated with higher earnings.

Figure 2

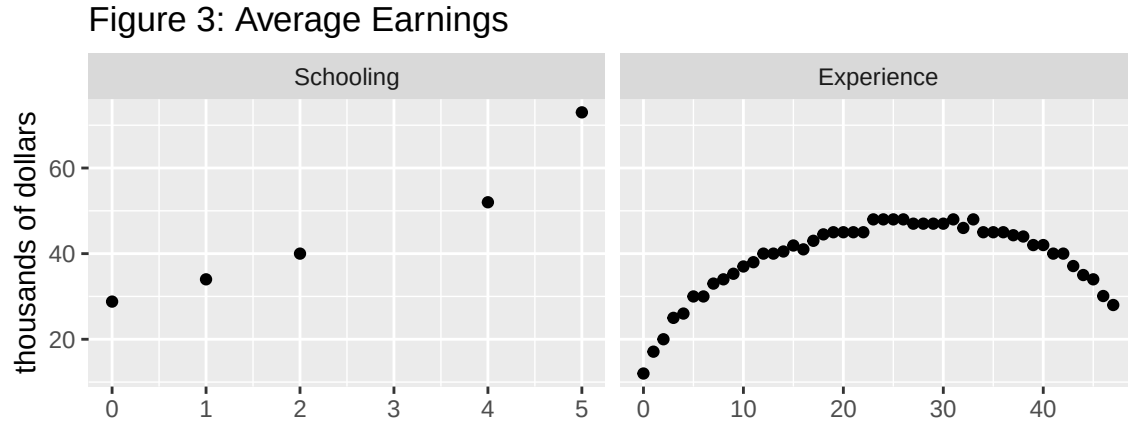


I will expand on this analysis by using two-sample t-tests to determine whether difference in means between groups is statistically significant. Specifically, my null hypothesis in these t-tests is that the difference in means for the high relative to the low groups is less than or equal to zero. Relatedly, my alternative hypothesis is that the differences in means are greater than zero. The results from these t-tests are shown in Table 4 below. Based on these t-tests, I can reject the null hypotheses: the findings support my previous observations that education and experience are positively related to earnings.

Notably, this mode of analysis is limited, as are my findings. First, grouping my sample into two bins (high or low) conditional on education or experience, may obscure some important trends. Second, my analyses have not taken any other variables into account (including each other). Thus, my findings may suffer from omitted variable bias. I can address the first of these concerns by plotting the average income for each level of schooling or experience, as shown in Figure 3 below. For schooling, average earnings increase with each additional year. Moreover, there are especially large gains from finishing college or graduate school. For experience, earnings also seem to increase with each additional year; however, this relationship flattens out after about twenty years. In addition, earnings begin to decrease after about thirty years. This actually makes sense if you consider that more work experience is correlated with age and there may be some productivity losses due to old age. This hump-shaped relationship was not visible from the previous analysis.

Table 4: Difference in Earnings Grouped by Schooling or Experience

	Mean Income, High	Mean Income, Low	T Statistic	P-Value
Schooling	65.26825	34.85263	308.3052	0
Experience	60.76971	47.26318	118.7392	0



Finally, I can extend the analysis shown in Figure 3 by regressing earnings on schooling and experience, respectively. The results from estimating these regressions are shown in Table 5. The results are in-line with my previous findings. An additional year of schooling is associated with roughly an additional ten thousand dollars while an additional year of experience is associated with an additional 500 dollars. Notably, the specification used for experience does not account for the hump-shape seen in Figure 3.

Table 5:

	<i>Dependent variable:</i>	
	Income	
	1	2
	(1)	(2)
Schooling	10.805*** (0.028)	
Experience		0.571*** (0.005)
Constant	32.323*** (0.078)	41.496*** (0.117)
Observations	1,174,328	1,174,328
<i>Note:</i>	*p<0.1; **p<0.05; ***p<0.01	

Conclusion

Basic economic theory suggests that more education or experience imparts skills and knowledge to workers that increases their productivity and therefore their wages. My analysis of 2017 ACS data, supports this theory. I find that earnings are positively related to education and positively related to work experience during roughly the first twenty-five years after school completion. Beyond that, earnings are declining in work experience possibly due to, for example, declines in health or stale skills. Notably, my analysis just shows the correlation between earnings, education, and experience; these results cannot be interpreted causally. In particular, the analysis does not include important possible confounders such as gender or ability, which might jointly affect both earnings and years of schooling. In future work, I hope to further explore the relationships between earnings with education and experience jointly while controlling for relevant covariates. Also, I will further examine the seemingly decreasing marginal returns of experience exhibited in Figure 3 at high levels of experience.

References

Data:

Steven Ruggles, Sarah Flood, Ronald Goeken, Josiah Grover, Erin Meyer, Jose Pacas and Matthew Sobek. IPUMS USA: Version 9.0 [dataset]. Minneapolis, MN: IPUMS, 2019. <https://doi.org/10.18128/D010.V9.0>

R Packages

ggplot2, readxl, dplyr, tidyr, ggthemes, stargazer, magrittr, gmodels, gridExtra, kableExtra, broom, ggpubr, Hmisc, weights